1- Masala

Berilgan

l=1,5 m

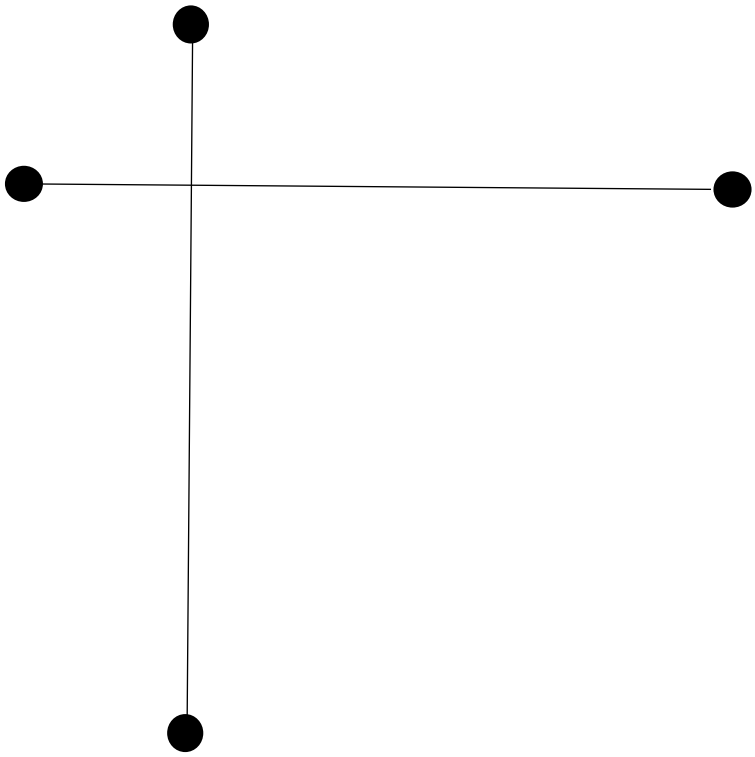
$m_1 = 1\text{kg}$

$m_2 = 2\text{ kg}$

$\boldsymbol{g=10\ ^m/s^2}$

$l_1 = 0{,}3\ m$

$l_2 = 1{,}2\ m$



v_{min} =?

Energiyaning saqalanish qonuniga ko’ra

$$\frac{m_1\omega^2 l_1}{2} + \frac{m_2\omega^2 l_2}{2} + m_1 gl = m_2 gl + m_1 g(l_2 - l_1)$$

$$0{,}15\omega^2 + 1{,}2\omega^2 + 15 = 30 + 10 * 0{,}9$$

$$1{,}35\omega^2 = 24$$

$\omega = \frac{8\sqrt{5}}{3}$ Javob : $\omega = \frac{8\sqrt{5}}{3}$

2- Masala

Berilgan

$v = 2\ m/s$

$d = 15\ sm$

$F = 10\ sm$

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v_t =?

$$\frac{1}{F} = \frac{1}{d} + \frac{1}{f}$$

$$f = \frac{dF}{d-F}$$

f =30sm

$$k = \frac{f}{d} = \frac{v_t}{v}$$

$$v_t = \frac{f}{d} * v = 2v = 4\frac{m}{s}$$

javob: $v_t = 4\frac{m}{s}$

3- Masala

Berilgan $v = 10^6 m/s$ $r = 5\ sm$ $e = 1{,}6 \cdot 10^{-19} c$ $m = 9{,}1 \cdot 10^{-31} kg$	harakatlanib kelayotgan elektronlarning kinetic energiyasi sharcha sirtiga yig'ilgan elektronlarning potensial energiyasiga tenglashguncha elektronlar oqadi.
N=?	

$$\frac{mv^2}{2} = \frac{k\,Ne^2}{R}$$

$$N = \frac{mv^2R}{2ke^2}$$

$$N = \frac{9{,}1 \cdot 10^{-31} kg * \left(\frac{10^6 m}{s}\right)^2 * 5 * 10^{-2} m}{2 * 9 * 10^9 \left[\frac{N * m^2}{kg^2}\right] (1{,}6 \cdot 10^{-19} c)^2} \approx 10^8\ ta$$

Javob: $N \approx 10^8\ ta$